Analyzing User Satisfaction in Deepseek Reviews Using RoBERTa and BERTopic

Satria Imawan [a,1,*], Yahya Nur Ifriza [b,2]

[a] Department of Computer Science, Faculty of Mathematics and Natural Sciences, Universitas Negeri Semarang, Semarang, Indonesia
*Corresponding author: satriaimawan23@students.unnes.ac.id

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ABSTRACT

The rapid growth of generative AI chatbots has intensified market competition, with Deepseek emerging as a notable case by achieving 20 million daily active users within 20 days of launch. This study analyzes user satisfaction of the Deepseek application through integrated sentiment analysis using RoBERTa and topic modeling using BERTopic on user reviews from Google Play Store. The research employs a quantitative descriptive-analytical approach with systematic stages: data collection, preprocessing, sentiment analysis, and topic modeling to identify factors influencing user satisfaction and dissatisfaction. Results reveal a polarized sentiment distribution with 52.6% positive, 38.9% negative, and 8.5% neutral, indicating a "positive but fragile" satisfaction condition. Topic modeling identified 32 topics consolidated into 11 topics based on predetermined criteria, with "Kualitas Aplikasi" (Application Quality) as the largest topic (1,256 reviews, 95.1% positive) and "Kualitas Server" (Server Quality) as the main pain point (1,036 reviews, 65.5% negative). Topic evolution analysis revealed stable patterns in application quality (38-42% consistently), positive increases in "Kualitas Jawaban" (Answer Quality) (8% to 16%), concerning negative increases in "Stabilitas & Bug" (Stability & Bugs) (8% to 18%), and positive decreases in account access issues (5% to 0%). User satisfaction is built upon three main factors: fundamental product quality, functional utility, and inclusive business model, while dissatisfaction centers on server infrastructure issues, application stability, and response performance. This research confirms that although DeepSeek has strong product-market fit, server quality issues create significant risks to long-term user retention requiring highest priority handling.

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1 Introduction

The rapid evolution of artificial intelligence has transformed how humans interact with technology, with AI-powered chatbots emerging as one of the most impactful applications. These conversational agents leverage natural language processing (NLP) to understand and respond to user queries, serving millions of users across education, business, and personal productivity domains. The global chatbot market, valued at USD 7.76 billion in 2024, is projected to reach USD 27.29 billion by 2030, reflecting a compound annual growth rate (CAGR) of 23.3% (Grand View Research, 2024). This exponential growth underscores the critical role of chatbots in modern digital ecosystems.

Within this competitive landscape, DeepSeek has emerged as a notable case study. Developed by Hangzhou DeepSeek Artificial Intelligence, this generative AI chatbot achieved remarkable adoption, reaching 20 million daily active users within just 20 days of launching its DeepSeek-R1 model in early

2025. Independent benchmarks revealed that DeepSeek-R1's free version matched or even surpassed the performance of premium models from established players like ChatGPT Plus, Gemini Advanced, and Claude Pro (Zhang et al., 2025). This success challenges conventional assumptions about market dominance and raises important questions about the factors driving user satisfaction in the hyper-competitive AI chatbot market.

Understanding user satisfaction has become a strategic imperative in this context. Research demonstrates that satisfied users are five times more likely to continue using an application and three times more likely to recommend it to others (Anderson & Sullivan, 1993). In markets with low switching costs, such as AI chatbots, where users can easily transition between alternatives, satisfaction directly impacts user retention and long-term viability. The Technology Acceptance Model (TAM) emphasizes that perceived usefulness and perceived ease of use are fundamental determinants of user attitudes and continued usage intentions (Davis, 1989). However, in fast-evolving applications like DeepSeek, understanding what specifically satisfies or dissatisfies users requires granular, dynamic analysis.

User reviews on platforms like Google Play Store provide a rich, authentic source of feedback that reflects real-world experiences. Unlike structured surveys that may suffer from response bias, organic reviews capture spontaneous user opinions about specific features, technical issues, and overall experiences. Approximately 79% of mobile app users consult ratings and reviews before downloading an application (Shepherd, 2024), highlighting the influence of user-generated content on acquisition and perception. For developers, these reviews contain actionable insights about feature appreciation, pain points, bugs, feature requests, competitive comparisons, and perception evolution across app versions. However, the challenge lies in the volume and unstructured nature of this data. For a rapidly growing application like DeepSeek, manual analysis of thousands of reviews is impractical and cannot provide real-time insights.

Previous research has demonstrated the value of integrating sentiment analysis with topic modeling for understanding user feedback. Luiz et al. (2018) showed that aggregate sentiment analysis alone, simply knowing that 60% of reviews are positive and 40% negative, fails to provide actionable insights about why users feel satisfied or dissatisfied. Yang et al. (2021) reinforced this by demonstrating that combining sentiment analysis with topic modeling creates interpretive synergy: sentiment indicates whether users are satisfied, while topics explain why these reactions occur. However, most existing studies treat reviews as static collections, failing to account for temporal dynamics and product evolution. In reality, each version of an application introduces changes such as bug fixes, new features, performance optimizations, or unintentional regressions, where user response to these changes is highly dynamic and contextual.

This research addresses this gap by proposing a dynamic, version-aware analytical framework. Rather than analyzing all reviews as a single corpus, we segment reviews based on both sentiment category (positive, negative, neutral) and application version. This dual segmentation enables topic evolution analysis that is far richer and more actionable. For instance, if "server issues" dominated complaints in version 1.0, we can track whether this topic's frequency decreased in subsequent versions (indicating effective fixes) or increased (indicating deteriorating problems). Similarly, new topics emerging in specific versions can indicate bugs introduced by updates or features gaining user attention. This evolution analysis provides valuable feedback loops for iterative, data-driven product development.

We employ state-of-the-art NLP models for this analysis: RoBERTa for sentiment classification and BERTopic for topic modeling. RoBERTa, an optimized variant of BERT, has demonstrated superior performance in sentiment analysis tasks compared to traditional baselines (Liu et al., 2019; Guo, 2024). BERTopic leverages transformer-based embeddings to generate more coherent and stable topics than traditional methods like Latent Dirichlet Allocation (LDA), particularly for short texts like app reviews (Grootendorst, 2022; An et al., 2023).

By analyzing user reviews of DeepSeek from Google Play Store, this study aims to: (1) classify sentiment to assess overall user satisfaction, (2) identify specific topics discussed within each sentiment-version combination, (3) analyze topic evolution across application versions to detect

emerging issues and verify fix effectiveness, and (4) identify factors influencing user satisfaction and dissatisfaction. These insights will provide actionable recommendations for developers to prioritize improvements, enhance user satisfaction, and maintain competitive advantage in the rapidly evolving AI chatbot market.

2 Method

2.1 Data Collection and Preparation

Data collection was conducted through web scraping using the google-play-scraper Python library to extract user reviews from the official Deepseek application page on Google Play Store. The target population comprised reviews from Indonesian users (country = 'id') posted between January and September 2025. Each review entry was extracted with essential attributes including review content (text), rating score (1-5 stars), posting date, and application version at the time of review submission. The collected raw data was stored in CSV format for subsequent processing and analysis. Initial data exploration examined review length distribution to understand dataset characteristics before preprocessing.

2.2 Preprocessing

Data preprocessing was conducted through three sequential stages to ensure data quality and consistency. First, case folding converted all characters to lowercase to eliminate capitalization variations and reduce vocabulary size. Second, cleansing removed noise elements including HTML tags, URLs, control characters (newlines, tabs), numbers, punctuation marks, and special characters using regular expressions. Emojis and emoticons were converted to textual representations using the emoji Python library to preserve emotional context rather than being discarded. Third, normalization standardized informal language variations, slang terms, abbreviations, and typos using a pre-compiled normalization dictionary containing key-value pairs mapping non-standard terms to their standard Indonesian equivalents. Each token in the preprocessed text was checked against this dictionary and replaced if a mapping existed. This preprocessing pipeline ensured that semantically equivalent terms would be recognized as identical by subsequent models.

2.3 Exploratory Data Analysis

During the Exploratory Data Analysis (EDA) phase, frequency analysis was performed to identify the most frequently occurring tokens within the review collection. The results of this analysis provide initial insights into the dominant terms reflecting user experiences, ranging from positive descriptors such as 'stable' or 'easy' to negative ones like 'lag' or 'error.' As a visual aid, a word cloud was generated to intuitively present the frequency distribution, allowing readers to immediately perceive the word distribution and its relevance priority at a glance. Furthermore, the distribution of review lengths was analyzed to gain a deeper understanding of the dataset's characteristics, enabling the determination of more appropriate preprocessing stages.

2.4 Sentiment Classification with RoBERTa

Sentiment classification was performed using a pre-trained RoBERTa model specifically fine-tuned for Indonesian text sentiment analysis (w11wo/indonesian-roberta-base-sentiment-classifier). This model was loaded from the Hugging Face Model Hub and configured through a sentiment analysis pipeline that processes each review text as input and outputs one of three sentiment categories: positive, negative, or neutral. The RoBERTa model architecture employs a robustly optimized BERT pretraining approach that removes the Next Sentence Prediction task, uses dynamic masking, and trains with larger batches compared to original BERT, resulting in improved performance on sentiment classification tasks. Model performance evaluation was conducted using a manually labeled validation subset to calculate accuracy, precision, recall, and F1-score metrics, ensuring reliability of sentiment predictions before using them for subsequent analysis.

2.5 Evaluation of the RoBERTa Model

Evaluating the performance of the RoBERTa model in classifying the sentiments of Deepseek application user reviews is a crucial step in ensuring the validity and reliability of the sentiment analysis results used in this study. The model employed is the RoBERTa Base Sentiment Classifier

(w11wo/indonesian-roberta-base-sentiment-classifier), which has been specifically fine-tuned for Indonesian sentiment analysis. This evaluation was conducted using a manually labeled subset of data as the ground truth to compare the model's predictions against the actual labels. The model's performance was measured using several standard metrics for multi-class classification, including accuracy, precision, recall, and F1-score.

2.6 Topic Modeling with BERTopic

Topic modeling was implemented using BERTopic, a framework that combines transformer-based embeddings, dimensionality reduction, density-based clustering, and class-based TF-IDF representation. The modeling pipeline consists of five sequential stages. First, each review text is converted into dense semantic embeddings using the SentenceTransformer model (paraphrase-multilingual-MiniLM-L12-v2), which supports Indonesian language and produces 384-dimensional vectors capturing contextual word meanings. Second, UMAP (Uniform Manifold Approximation and Projection) reduces the high-dimensional embeddings to 5-15 dimensions while preserving both local neighborhood structure and global data topology. The n_neighbors parameter controls the balance between local and global structure preservation, while min_dist parameter determines the minimum distance between points in the reduced space. Third, HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) performs clustering by identifying high-density regions and automatically determining the number of clusters without requiring pre-specification. Documents not fitting well into any cluster are labeled as outliers or noise. Fourth, class-based TF-IDF (c-TF-IDF) generates topic representations by concatenating all documents within each cluster into a single "super-document" and calculating term importance within the cluster relative to other clusters, producing ranked keyword lists that characterize each topic. Fifth, topic consolidation merges semantically similar topics based on keyword overlap analysis and manual thematic coherence assessment to produce interpretable main topics representing key discussion themes.

2.7 Evaluation of the BERTopic Model

Evaluating the topic modeling model is a crucial stage to ensure that the topics generated by BERTopic are of high quality, meaningful, and interpretable. Unlike supervised learning models, which can be evaluated using standard metrics such as accuracy or F1-score, topic modelling as an unsupervised learning method requires an evaluation approach that focuses more on semantic quality and the internal data structure using topic coherence and topic diversity. In this study, the evaluation of the BERTopic model was conducted to ensure the validity and reliability of the topic modeling results.

2.8 Analysis of Results

The analysis of results is the most pivotal stage of this research. It evaluates user satisfaction by synthesizing sentiment analysis and topic modeling from Deepseek review data. The process starts with a sentiment distribution analysis to provide a general overview of user satisfaction levels. This is followed by a frequency analysis to delve into the specific drivers of each sentiment. For topic modeling, the analysis begins by inspecting word representations, with subsequent topic consolidation if necessary. The results are then cross-referenced with sentiment labels to determine the key factors behind user satisfaction and dissatisfaction. Lastly, a topic evolution analysis is conducted to track thematic changes across application versions.

3 Results and Discussion

3.1 Data Collection Results

In this study, user reviews of the Deepseek application were retrieved using web scraping techniques implemented in Python via the google_play_scraper library. The data acquisition process yielded a dataset consisting of 3,613 rows and 11 columns, namely: *reviewId*, *userName*, *userImage*, *content*, *score*, *thumbsUpCount*, *reviewCreatedVersion*, *at*, *replyContent*, *repliedAt*, and *appVersion*. The collected reviews span a period from January to September 2025.

Table 1. Web scraping results

Idx	ReviewID	Username	Content	At	Score	Appversion
1.	b8cb155b-68a8-4521-9d7a-645247c87e72	Pengguna Google	Aplikasi 👍	2025-09-17 01:30:42	5	1.3.1
…	…	…	…	…	…	…
3613	a4bda92d-5876-4905-93f0-4d2df12f03ce	Pengguna Google	Hi, im the FIRST USER	2025-01-14 13:37:27	5	1.0.1

3.2 Dataset Preparation Results

Dataset preparation is a critical phase conducted prior to the preprocessing stage. This stage focuses on understanding the general characteristics of the data. The first step involves categorizing the length of each review into three distinct categories: micro-text (less than 10 words), short text (between 10 and 50 words), and long text (more than 50 words). As illustrated in Figure 1, the distribution results indicate that the majority of reviews are categorized as micro-text. This suggests that users tend to provide brief and concise comments without offering additional explanations. This prevalence of short comments presents a significant challenge for the subsequent analysis stages due to the limited contextual information provided in the dataset.

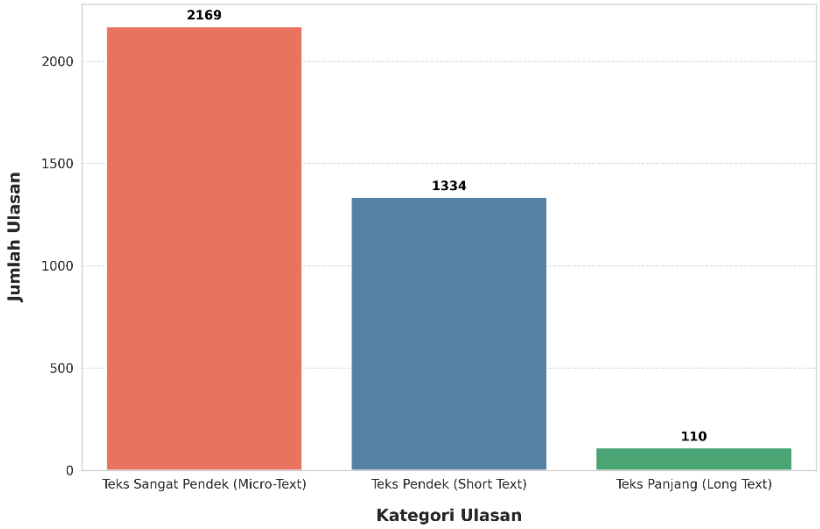

Figure 1. Distribution of reviews by text length category

3.3 Preprocessing Results

In the preprocessing stage, three techniques were employed: case folding, cleansing, and normalization to clean, standardize, and enhance the quality of the text data. These steps are essential to ensure optimal performance and accuracy in both sentiment classification and topic modeling results. An example of preprocessing on Deepseek user review data can be seen in Table 2.

Table 2. Preprocessing results

Content	Case folding	Cleansing	Normalization
Lag terozzzz 😠, Tiap akhir obrolan pasti begitu.	lag terozzzz wajah marah, tiap akhir obrolan pasti begitu.	lag terozzzz wajah marah tiap akhir obrolan pasti begitu	lag terus wajah marah tiap akhir obrolan pasti begitu

3.4 Exploratory Data Analysis Results

Exploratory Data Analysis (EDA) in this study was conducted in two stages: word cloud and frequency analysis to identify initial patterns in data. A word cloud is a visual representation of words based on their frequency of occurrence; the higher the frequency, the larger the word appears in the visualization. As illustrated in Figure 2, words such as “bagus”, “ai”, “server”, “sibuk”, and “Deepseek” are visually prominent, indicating a strong duality of sentiments within the dataset.

Figure 1. Visual representation of word frequency through a word cloud

For a more measurable analysis, the next stage involves word frequency analysis to identify the most commonly occurring terms. Based on the results of the frequency analysis, the 20 most frequent words in DeepSeek application reviews are presented in Table 3. The analysis revealed a highly polarized sentiment landscape. On the positive side, words like "bagus" (954 occurrences), "jempol" (thumbs up, 270), "membantu" (helpful, 230), and "keren" (165) dominated, indicating strong appreciation for application quality. However, negative words were equally prominent, with "server" (644), "sibuk" (576), and "error" (155) appearing frequently. The phrase "server sibuk" (busy server) emerged as the most common complaint. Notably, "chatgpt" (160) also appeared frequently, indicating active comparisons with DeepSeek's main competitor.

Table 3. Word frequency

No	Word	Frequency	No	Word	Frequency
1	Bagus (Good)	954	11	Fitur (Feature)	165
2	Ai	843	12	Keren (Cool)	165
3	Server	644	13	Chatgpt	160
4	Sibuk (Busy)	576	14	Bintang (Star)	159
5	Deepseek	361	15	Mantap (Excellent)	156
6	Banget (Very)	301	16	Error	155
7	Tolong (Please)	284	17	Gratis (Free)	154
8	Jempol (Thumbs up)	270	18	Servernya (The server)	140
9	Membantu (Helpful)	230	19	Perbaiki (Repair)	130
10	Chat	207	20	Udah (Already)	129

3.5 Sentiment Analysis

3.5.1 Sentiment Analysis Results

Sentiment analysis was performed to classify all reviews into three polarities: positive, neutral, and negative. This analysis serves as a crucial step in assessing the overall level of user satisfaction with the application. To ensure the validity of the sentiment classification results, the model was evaluated using standard metrics, including accuracy, precision, recall, and the F1-score. The evaluation results

of the RoBERTa model on the Deepseek user review dataset, as illustrated in Table 4, demonstrate excellent overall performance.

Table 4. Performance evaluation of the RoBERTa model

No	Metric	Score	Range
1.	Accuracy	0,9723	The higher, the better (0-1)
2.	Precision	0,9733	The higher, the better (0-1)
3.	Recall	0,9723	The higher, the better (0-1)
4.	F1-score	0,9726	The higher, the better (0-1)

The sentiment distribution of Deepseek application reviews is presented in Figure 3. The chart indicates that 52.6% of the reviews express a positive sentiment, 38.9% exhibit a negative sentiment, and the remaining 8.5% reflect a neutral sentiment.

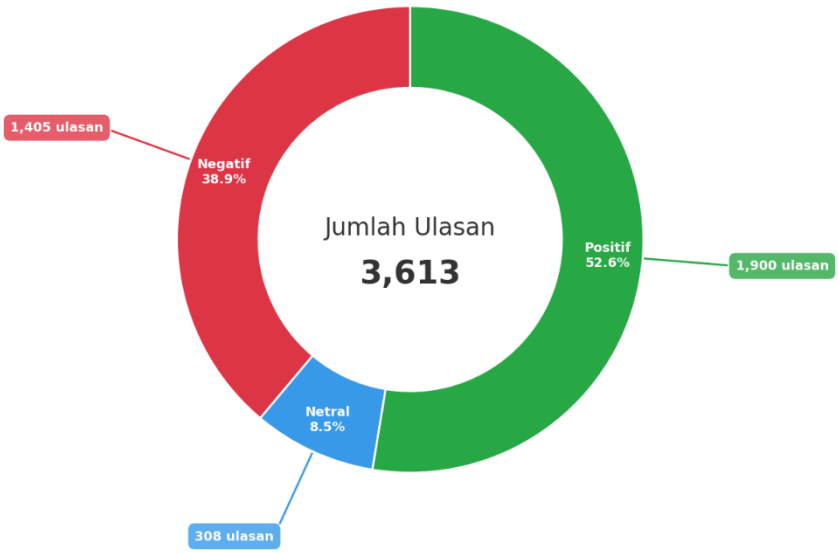

Figure 3. Sentiment distribution of Deepseek application reviews

The results of the frequency analysis for positive sentiments, as shown in Figure 4, reveal several interesting findings. The words 'bagus' (good, 661) and 'ai' (470) dominate user conversations, followed by other expressions of appreciation such as 'jempol' (220) and 'membantu' (helpful, 212).

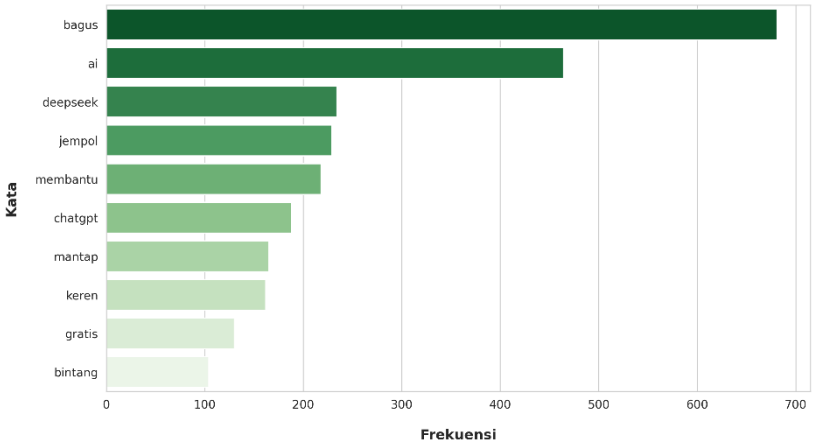

Figure 4. Frequency analysis of positive sentiment reviews

Meanwhile, the frequency analysis results for negative sentiments in Figure 5 indicate that system quality issues are the primary source of user dissatisfaction. This is characterized by the dominance

of terms such as “server” (531) and “sibuk” (busy, 480), suggesting that the phrase “server sibuk” is the most frequent complaint. These technical issues are further evidenced by the occurrence of the word “error” (131) and urgent user appeals such as “tolong” (please, 182). Interestingly, the terms “ai” (326) and “bagus” (good, 285) also appear, likely occurring within phrases such as “AI-nya bagus, tapi servernya sibuk” (The AI is good, but the servers are busy.).

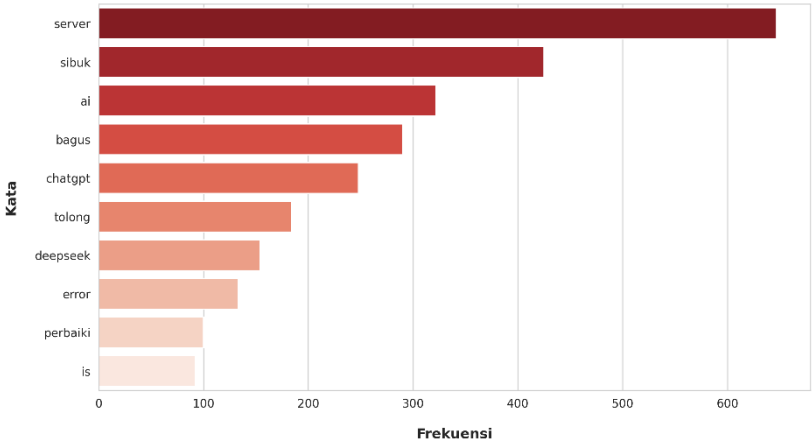

Figure 5. Frequency analysis of negative sentiment reviews

3.5.2 Sentiment Analysis Discussion and Interpretation

Sentiment distribution revealed a polarized landscape: 52.6% positive, 38.9% negative, and 8.5% neutral sentiments. While positive sentiment dominates, the relatively narrow margin (13.7%) indicates a "positive but fragile" satisfaction state. Frequency analysis of positive sentiments highlighted keywords like "bagus" (good, 661), "membantu" (helpful, 212), and "gratis" (free, 126), reflecting appreciation for product quality, usefulness, and free access model. Conversely, negative sentiments concentrated on "server" (531) and "sibuk" (busy, 480), identifying infrastructure as the primary pain point. This polarization aligns with the Expectation Confirmation Theory (Oliver, 1980; Bhattacherjee, 2001), where users experiencing positive disconfirmation appreciate the quality while those facing server issues experience negative disconfirmation that amplifies frustration. The findings demonstrate a critical tension between strong product-market fit and infrastructure limitations.

3.6 Topic Modeling

3.6.1 Topic Modeling Results

Based on the results of the topic modeling conducted using BERTopic on a dataset of 3,613 Deepseek user reviews, the model generated a total of 34 topics, as detailed in Table 5.

Table 5. Topic information from BERTopic results

Topic	Word Representation	Total
-1	['yang', 'bagus', 'dan', 'ai', 'saya', 'chatgpt', 'di', 'ini', 'overload', 'sangat']	507
…	…	…
32	['mantap mantap', 'mantap', 'mantap untuk', 'recomended', 'pro', 'untuk yang', 'membantu mantap', 'joss', 'biasa mantap', 'ai mantap']	35

Due to their specific characteristics and the substantial amount of data categorized as outlier topics, these data points were subjected to a subsequent round of topic modeling using BERTopic. The results of this secondary topic modeling process are presented in Table 6.

Table 6. Topic information from BERTopic results after outlier handling

Topic	Word Representation	Total
0	['server', 'sibuk', 'server sibuk', 'sering', 'selalu', 'is', 'server is', 'tolong', 'tapi', 'nya']	1036
…	…	…
31	['membantu sangat', 'sangat membantu', 'membantu', 'sangat', 'very', 'bekerja', 'berguna dan', 'amazing', 'dan membantu', 'membantu dalam']	41

Following the completion of the outlier handling process, evaluating the quality of the generated topics is a crucial step to ensure the validity and reliability of the topic modeling results. This evaluation is conducted using topic coherence and topic diversity metrics, as presented in Table 7.

The topic coherence score of 0.5573 indicates a good level of semantic relatedness among keywords within each topic. This value falls within an acceptable range for topic modeling on short texts like app reviews, where coherence scores tend to be lower than for longer documents due to limited word co-occurrence within single documents. The score of 0.5573 demonstrates that the keywords representing each topic have sufficiently strong semantic relationships, meaning the topics can be well interpreted. The topic diversity score of 0.7182 is particularly strong, indicating clear semantic differences between topics without excessive overlap. High diversity ensures that each topic represents a distinct aspect of user reviews rather than minor variations of the same theme. Together, these evaluation metrics provide strong confidence that the identified topics offer meaningful and valid insights into factors influencing user satisfaction and dissatisfaction.

Table 7. Performance evaluation of the BERTopic model

No	Metric	Score	Range
1.	Topic coherence	0,5573	The higher, the better (0-1)
2.	Topic diversity	0,7182	The higher, the better (0-1)

Furthermore, to facilitate interpretation and enhance analytical effectiveness, the 32 topics generated by BERTopic were consolidated into 11 main topics: “Kualitas Server”, “Kualitas Aplikasi”, “Kecepatan Respon Jawaban”, “Stabilitas Aplikasi & Bug”, “Kualitas Jawaban”, “Manfaat & Kegunaan”, “Akses Akun & Otentifikasi”, “Perbandingan Kompetitor”, “Saran”, “Harapan Pengguna”, and “Monetisasi”. This consolidation, based on thematic similarities and conversational context, is detailed in Table 8. Moreover, the results of this consolidation from 32 topics into 11 primary categories can be observed in Table 9.

Table 8. Criteria for consolidating 32 topics into 11 main topics

No	Topic	Definition	Inclusion Criteria
1.	Kualitas Server/Server Quality	Topics discussing all aspects related to server infrastructure and application service availability.	- Keywords: "server", "sibuk", "overload", "down"
2.	Kualitas Aplikasi/Application Quality	Topics discussing the general assessment of overall application quality, including user experience, design, and core functionality.	- Keywords: "bagus", "keren", "mantap", "aplikasi", "luar biasa", "terbaik"
3.	Kecepatan Respon Jawaban/Response Speed	Topics discussing the application's speed in processing requests and delivering responses to users.	- Keywords: "cepat", "lambat", "lama", "respon", "loading", "lemot", "gacor"
4.	Stabilitas Aplikasi & Bug/Stability & Bugs	Topics discussing technical issues related to application stability, including bugs, errors, crashes, and specific functional glitches.	- Keywords: "error", "bug", "crash", "tidak bisa", "gagal", "upload", "gambar"

5.	Kualitas Jawaban/ Answer Quality	Topics discussing the quality, accuracy, relevance, and contextualization of the responses provided by the AI, including its multilingual capabilities.	- Keywords: "akurat", "bodoh", "jawaban", "salah", "bahasa", "indonesia", "inggris", "china"
6.	Manfaat & Kegunaan/Benefits & Usefulness	Topics discussing the utility and benefits of the application in assisting users to complete their tasks or fulfill their needs.	- Keywords: "membantu", "berguna", "mudah", "informasi"
7.	Akses Akun & Otentikasi/Account Access	Topics discussing issues related to the registration process, login, account verification, and user authentication.	- Keywords: "login", "akun", "daftar", "kode", "email", "google", "verifikasi"
8.	Perbandingan Kompetitor/ Competitor Comparison	Topics discussing comparisons between DeepSeek and other generative AI chatbot applications.	- Keywords: "chatgpt", "gemini", "lebih", "dari chatgpt", "lebih baik"
9.	Saran/ Suggestions	Topics containing requests for new features, proposals for improvement, or development suggestions from users.	- Keywords: "tambahkan", "fitur", "suara", "voice", "update", "data", "terbaru", "2024", "2025"
10.	Harapan Pengguna/ User Expectations	Topics expressing expectations for future application improvements and development.	- Keywords: "semoga", "lebih baik", "lebih bagus", "makin", "kedepan", "berkembang"
11.	Monetisasi/ Monetization	Topics discussing the application's business model, pricing policies, and the perceived economic value offered.	- Keywords: "gratis", "premium", "bayar", "murah"

Table 9. Topic information from BERTopic results after outlier handling

Topic	Word Representation	New Topic
0	['server', 'sibuk', 'server sibuk', 'sering', 'selalu', 'is', 'server is', 'tolong', 'tapi', 'nya']	1_Kualitas server
…	…	…
31	['membantu sangat', 'sangat membantu', 'membantu', 'sangat', 'very', 'bekerja', 'berguna dan', 'amazing', 'dan membantu', 'membantu dalam']	5_Manfaat & Kegunaan

3.6.2 Topic Distribution by Sentimen

After consolidating the 32 initial topics into 11 core themes, the subsequent step involves analyzing the topic distribution based on sentiment categories. This analysis aims to identify which topics drive satisfaction and which drive dissatisfaction, we analyzed topic distribution across sentiment categories. This analysis reveals whether each topic is predominantly discussed in positive, negative, or neutral contexts. The visualization of the review count distribution for each topic based on sentiment is presented in Figure 6.

The first pattern shows topics with predominantly positive sentiment (>90% positive), which represent the core satisfaction drivers identified in the data. "Kualitas Aplikasi" emerges as the largest topic with 1,256 reviews showing 95.1% positive sentiment, 4.9% negative sentiment, and 0% neutral sentiment. This overwhelming positivity confirms that users consistently appreciate the application's fundamental design, interface, and basic functionality. "Manfaat & Kegunaan" demonstrates even stronger unanimity with 170 reviews showing 100% positive sentiment and no

negative or neutral reviews, indicating universal appreciation for the application's practical value in meeting user needs. "Monetisasi" shows 52 reviews with 92.3% positive and only 7.7% negative sentiment, reflecting strong appreciation for the free access model without feature limitations.

The second pattern reveals topics with predominantly negative sentiment (>60% negative), representing critical pain points requiring priority attention. "Kualitas Aplikasi" stands out as the most problematic topic with 1,036 reviews distributed as 1.8% positive, 65.5% negative, and 32.6% neutral. The substantial neutral proportion alongside dominant negative sentiment suggests many users are uncertain or resigned about server reliability rather than actively condemning it. "Stabilitas Aplikasi & Bug" shows 193 reviews with extreme negativity at 88.6% negative and only 11.4% positive, indicating severe dissatisfaction with application stability and technical reliability. "Akses Akun & Otentifikasi" demonstrates complete negativity with 46 reviews showing 100% negative sentiment and no positive or neutral reviews, though this topic's relatively small size suggests it affects a limited user segment. "Kecepatan Respon Jawaban" shows 302 reviews with 66.6% negative and 33.4% positive sentiment, indicating that while processing speed issues affect the majority of users discussing this topic, a substantial minority actually appreciate the application's speed.

The third pattern includes topics with more balanced sentiment distribution (40-60% on either side), providing nuanced insights into divided user perceptions. "Perbandingan Kompetitor" shows 184 reviews almost perfectly split with 49.5% positive and 50.5% negative sentiment, indicating users are genuinely divided in their comparative assessments between DeepSeek and competitors like ChatGPT. This balance suggests DeepSeek occupies a contested competitive position—viewed as comparable or superior by half of comparing users, but still perceived as inferior by the other half. "Kualitas Jawaban" shows 300 reviews with 63.3% positive and 36.7% negative sentiment, indicating that while the majority appreciate the AI's response quality, a substantial minority remains dissatisfied. The remaining topics—"Saran", "Harapan Pengguna", and others show varied distributions but account for relatively small proportions of total reviews (each under 3%).

The magnitude and sentiment distribution of topics reveal important hierarchies in user priorities. The near-equal proportions of "Kualitas Aplikasi" (34.8% of all reviews) and "Kualitas Server" (28.7%) indicate these two aspects dominate user discourse. However, their opposite sentiment profiles 95.1% positive for application quality versus 65.5% negative for server quality create a stark contrast where appreciation for the application itself is almost matched by frustration with its availability. This juxtaposition represents an unusual situation where product strength is nearly equaled by infrastructure weakness in user commentary.

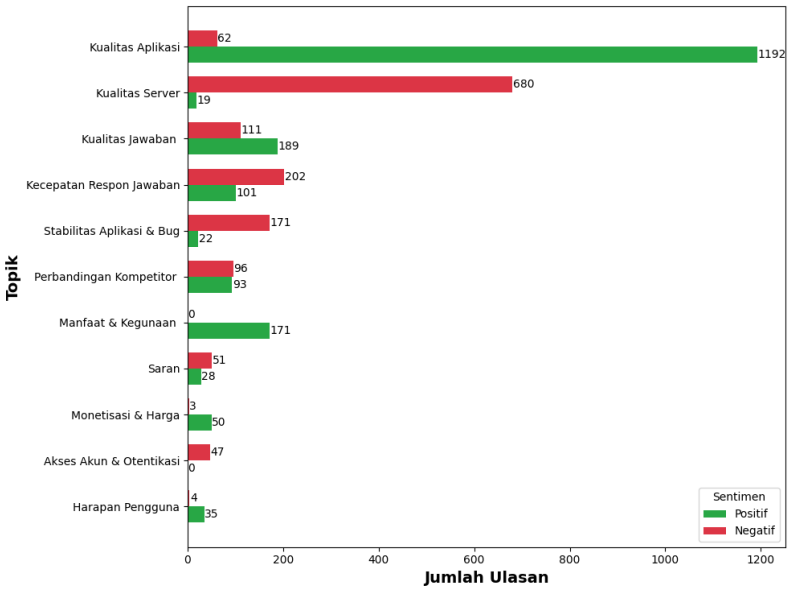

Figure 6. Count distribution for each topic based on sentiment

3.6.3 Topic Evolution Across Application Versions

To understand how user perceptions change with application updates, we analyzed topic evolution across major versions (1.0 to 1.3) separately for positive and negative sentiments. The visualization of topic evolution for reviews with positive sentiment is presented in Figure 7. Three main patterns emerged in positive reviews:

1. Stable Patterns: "Kualitas Aplikasi" consistently dominated with 38-42% proportion across all versions, demonstrating successful maintenance of fundamental quality standards through iterative development.
2. Positive Increase Patterns: "Kualitas Jawaban" increased from 8% (v1.0) to 16% (v1.3), and "Manfaat & Kegunaan" increased from 6% to 11%, indicating that improvements in these aspects are increasingly appreciated by users. The increasing appreciation for answer quality likely results from model fine-tuning, training data updates, or improvements in retrieval mechanisms.
3. Stable Minor Topics: Topics like "Monetisasi", "Perbandingan Kompetitor", "Harapan Pengguna", and "Saran" maintained relatively small proportions (1-4%) with minor fluctuations across versions, showing no clear trend.

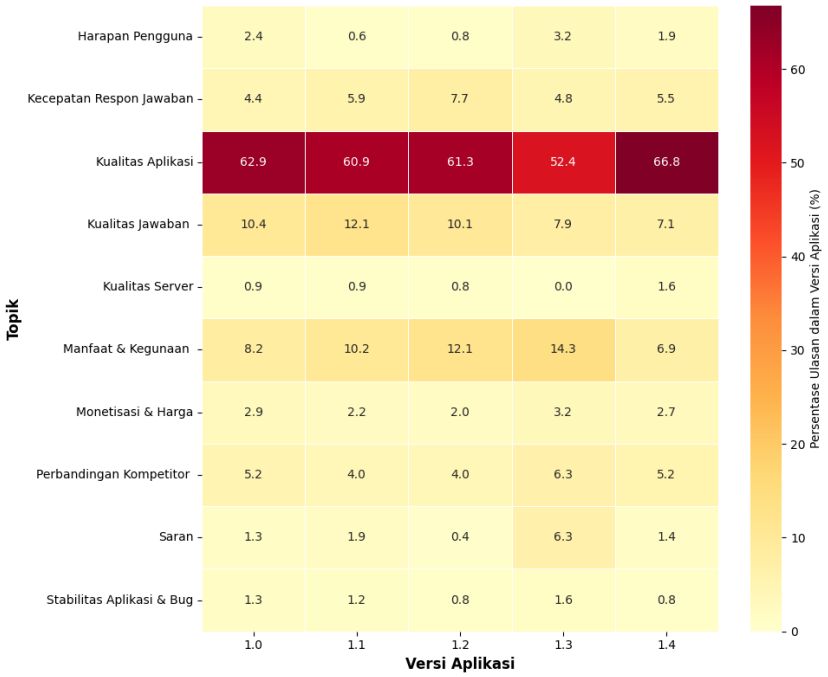

Figure 7. Evolution of topics across application versions for positive sentiments

The visualization of topic evolution for reviews with positive sentiment is presented in Figure 8. The negative review analysis revealed more concerning patterns:

1. Persistent Problems: "Kualitas Server" consistently dominated with 47-50% proportion across all versions, indicating that infrastructure problems remain unresolved despite potential capacity increases. This persistence suggests either that improvements are not proportional to demand growth or that problems lie in fundamental system architecture.
2. Concerning Increase: "Stabilitas Aplikasi & Bug" increased alarmingly from 8% (v1.0) to 18% (v1.3), indicating that each application update tends to introduce new bugs or cause regressions in previously working functions. This pattern suggests inadequate quality assurance and testing processes before release.
3. Successful Resolution: "Akses Akun & Otentifikasi" decreased dramatically from 5% (v1.0) to nearly 0% (v1.3), providing concrete evidence of successful problem identification and effective solution implementation by developers. This success demonstrates that when user

feedback is analyzed well and priorities are set appropriately, critical problems can be effectively resolved.

4. Stable Moderate Topics: "Response Speed" maintained stable proportion at 13-16% across versions, showing that speed issues remain a consistent complaint. "Answer Quality" in negative contexts remained stable at 8-10%, indicating that while many users appreciate answer quality (as seen in positive analysis), there remains a segment consistently finding problems with this aspect.

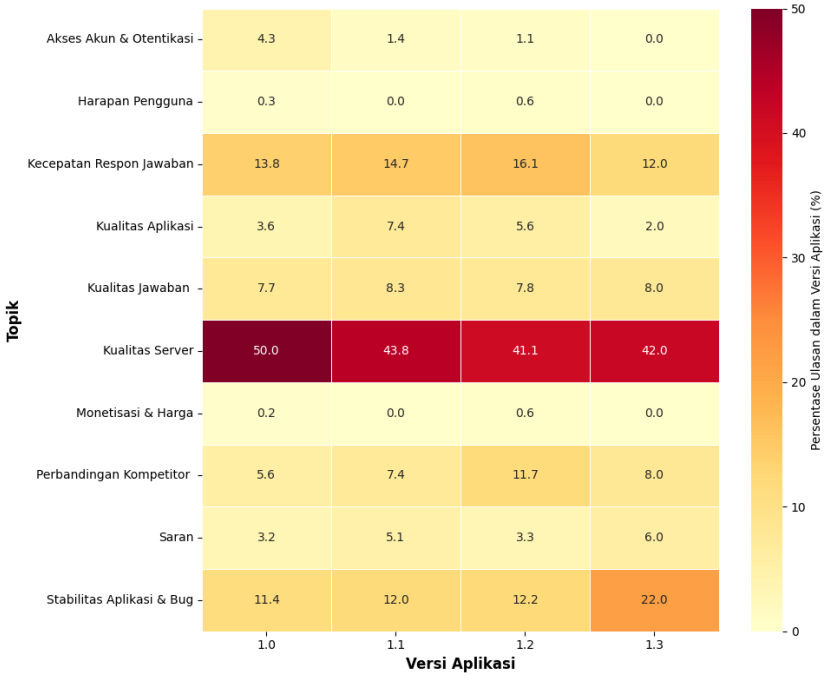

Figure 8. Evolution of topics across application versions for negative sentiments

3.6.4 Topic Modeling Discussion and Interpretation

3.6.4.1 Factors Driving User Satisfaction

User satisfaction is built upon three interconnected factor revealed by predominantly positive topics. First, fundamental product quality represented by "Kualitas Aplikasi" (1,256 reviews, 95.1% positive) demonstrates that users consistently appreciate the intuitive interface, attractive visual design, and reliable basic functionality. From the Technology Acceptance Model (TAM) perspective (Davis, 1989), this relates to perceived ease of use, indicating DeepSeek has successfully designed a user experience that is intuitive and requires minimal cognitive effort. Second, functional utility shown by "Manfaat & Kegunaan" (170 reviews, 100% positive) and "Kualitas Jawaban" (190 positive reviews, 63.3% positive) confirms strong perceived usefulness. Users explicitly state the application is "gratis", "membantu", and "mudah dipakai" in meeting various needs from education to work and research. The increasing trend of "Kualitas Jawaban" from 8% to 16% across versions indicates improvements in accuracy and relevance of AI responses are increasingly appreciated. Third, inclusive business model reflected in "Monetisasi" (52 reviews, 92.3% positive) where users repeatedly express appreciation for "gratis" access without feature limitations. This creates strong competitive differentiation and high value perception compared to competitors using freemium models with significant restrictions. Research by Kim et al. (2007) confirms that value for money perception significantly influences user satisfaction and loyalty, especially in competitive markets.

3.6.4.2 Factors Driving User Dissatisfaction

Three critical issues emerged as dissatisfaction sources. First, “Kualitas Server” (1,036 reviews, 65.5% negative) represents the most severe problem, attributed to scaling lag from exponential user growth (20 million daily active users in 20 days) exceeding infrastructure capacity. The computational complexity of generative AI models compounds this challenge, creating bottlenecks

during peak usage. This directly impacts service quality dimensions of availability and reliability in the DeLone and McLean model. Second, “Stabilitas Aplikasi & Bug” (193 reviews, 88.6% negative) and “Akses Akun & Otentifikasi” (46 reviews, 100% negative) reflect challenges in maintaining code quality during rapid development cycles. Third, “Kecepatan Respon Jawaban” (302 reviews, 66.6% negative) ties directly to infrastructure problems, where high server load increases latency. In conversational AI context, high latency severely degrades user experience despite quality answers.

3.6.4.3 Competitive Positioning

The “Perbandingan Kompetitor” topic (184 reviews, 49.5% positive vs 50.5% negative) reveals complex competitive dynamics. Half of users view Deepseek as comparable or superior to ChatGPT, validating its positioning as a credible alternative. However, the balanced distribution and increasing negative comparisons (3% to 5% across versions) indicate growing critical evaluation where Deepseek falls short, particularly in service stability.

Through disruptive innovation lens (Christensen, 2018), Deepseek targets price-sensitive segments with "good enough" AI quality plus free full access. However, in hyper-competitive markets with minimal switching barriers, quality gaps can trigger significant churn. Research by Li et al. (2020) shows that mobile apps rated 0.5 stars below competitors experience 30-50% download reduction.

3.6.4.4 Strategic Implications

The integrated analysis reveals that Deepseek has achieved strong product-market fit with appreciated quality, utility, and business model. However, persistent infrastructure issues create significant long-term retention risks. The 13.7% margin between positive and negative sentiments falls below the 70-80% satisfaction threshold recommended for strong loyalty (Anderson & Sullivan, 1993).

4 Conclusion

Sentiment analysis revealed a polarized user base with 52.6% positive, 38.9% negative, and 8.5% neutral sentiments, indicating a "positive but fragile" satisfaction state. The relatively narrow 13.7% margin between positive and negative sentiments suggests the application operates in a critical zone where persistent issues could rapidly shift overall satisfaction trajectory.

Topic modeling identified 11 core themes, with “Kualitas Aplikasi” (1,256 reviews, 95.1% positive) and “Kualitas Server” (1,036 reviews, 65.5% negative) emerging as dominant contrasting forces. This dichotomy reveals that while Deepseek has achieved strong product-market fit through quality fundamentals, infrastructure limitations create the most significant barrier to sustained user satisfaction. User satisfaction is built upon three interconnected factor: (1) fundamental product quality with intuitive interface and robust functionality, (2) functional utility providing tangible benefits for education, work, and information needs, and (3) inclusive business model offering full free access without feature restrictions. Conversely, user dissatisfaction centers on three critical categories: (1) infrastructure problems with server overload affecting 65.5% of negative reviews, (2) stability issues with bugs increasing from 8% to 18% across versions, and (3) performance concerns regarding response speed affecting 66.6% of complaints in this category.

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